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Day reference

Night query

Problem

We want a robot to recognize its location at night using a daytime map. This is hard because large illumination changes make many daytime-trained descriptors unstable at night.

Gap in existing work

Many VPR papers report strong benchmark performance, but it is still unclear which descriptors **transfer best to our specific setting**: day-to-night localization, training-free evaluation, and a small campus deployment scenario.

Research question

If we keep the pipeline simple and avoid retraining, which descriptor gives the most reliable nocturnal place recognition on both a controlled benchmark and our own campus dataset?

Hypothesis

Descriptor choice is the dominant factor in our current nocturnal VPR setup. We expect modern VPR descriptors to outperform older or more generic baselines on both GardensPoint and the strict campus dataset.

Descriptors evaluated

- **Older / generic**: SAD, AlexNet, NetVLAD
- **Local + global**: PatchNetVLAD
- **Modern VPR descriptors**: CosPlace, EigenPlaces, SALAD, SelaVPR++
- **Specialized spiking model**: VPRTempo

What these are

CosPlace / EigenPlaces learn strong global place descriptors. SALAD improves descriptor aggregation with DINOv2 features. SelaVPR++ adapts foundation models for efficient place recognition. VPRTempo is a fast temporally encoded spiking network rather than a universal drop-in descriptor.

Why this comparison matters

Our current phase focuses on **descriptor exploration first**: before adding CLAHE, sequence matching, or temporal filtering, we want to know which representation is strongest in our actual setting.

Methods still planned

CLAHE preprocessing, sequence matching, and temporal prediction remain future or next-phase improvements once the descriptor baseline is settled.

Dataset → Descriptor extraction → Cosine similarity → Matching + thresholding → AUC, R@100P, R@1/R@5/R@10

GardensPoint

200 day-right reference images and 200 night-right queries. This remains our main benchmark because it directly matches the **day-to-night** problem.

Campus dataset

50 day images and 64 night queries. In this talk we focus on the **strict image-level protocol**, which is the more conservative evaluation. All experiments are **training-free transfer tests** unless explicitly stated otherwise.

Descriptor	AUC	R@1	R@5	R@10
CosPlace	0.592	0.350	0.840	0.925
EigenPlaces	0.676	0.370	0.840	0.940
PatchNetVLAD	0.802	0.440	0.800	0.840
SALAD	**0.924**	**0.450**	**0.955**	**0.990**
SelaVPR++	**0.937**	**0.465**	0.950	**0.995**
VPRTempo	0.065	0.095	0.230	0.290

Examples: 1 correct (green) and 1 wrong (red) matches from S>=thresh

Takeaway

The strongest benchmark descriptors are now **SelaVPR++ and SALAD**. VPRTempo transfers very poorly with the official demo checkpoint.

Strict interpretation

No obvious leakage was found, but the most trustworthy evidence is still **R@1 / R@5 / R@10**. GardensPoint soft ground truth makes AUC and R@100P more lenient.

Day-night shift is dataset-dependent

The EDA shows that **GardensPoint and campus are not the same kind of night data**.

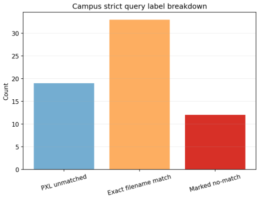
- **GardensPoint**: night images appear exposure-compensated or over-brightened, and are brighter on average.
- **Campus**: night images are darker and slightly lower in contrast than day images.

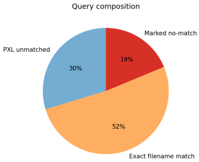
Strict campus is structurally harder

- 33 exact filename matches
- 12 marked no-match queries
- 19 phone images with no strict day counterpart

Why this matters

We keep **GardensPoint** as the controlled benchmark, but the EDA explains why the **campus transfer test** is harder: it combines a darker night domain with many hard no-match cases. It also tells us how to read experiments: benchmark gains do not automatically imply true dark-night robustness, and low campus precision may come from dataset structure



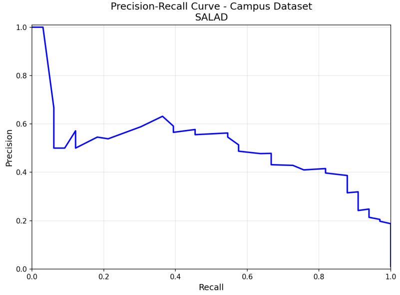


Descriptor	AUC	R@100P	R@1	R@5	R@10
CosPlace	0.470	0.030	0.727	1.000	1.000
HDC-DELF	0.488	0.030	0.788	0.939	0.970
SALAD	0.504	0.030	**0.818**	**1.000**	**1.000**
SelaVPR++	0.464	**0.061**	**0.909**	**1.000**	**1.000**
VPRTempo	0.064	0.000	0.182	0.515	0.606

Examples: 3 correct (green) and 5 wrong (red) matches from S>=thresh

Interpretation

Under strict 1-to-1 labels, **SelaVPR++ gives the strongest top-1 retrieval** while SALAD stays strong. No-match rejection remains difficult, and VPRTempo is not competitive in this transfer setting.



Key observation

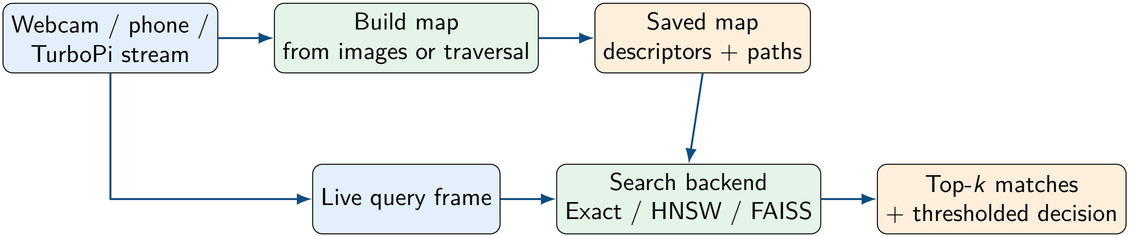
The campus setting is much harsher than GardensPoint because it mixes day-night change with viewpoint change, limited map coverage, and many hard no-match cases.

Interpretation

- **SelaVPR++** gives the best top-1 retrieval under the strict protocol.
- **SALAD** stays consistently strong and reaches perfect R@5 and R@10.
- **CosPlace** remains a strong practical baseline.
- **VPRTempo** is not competitive in this transfer setup.

Main lesson

Strong modern descriptors clearly help, but descriptor quality alone does not fully solve rejection and ambiguity in a small real campus deployment.



Current status

We already have an end-to-end live pipeline for map building and localization.
The next step is to connect the strongest descriptor choices and lightweight temporal ideas to the demo path.

1. Our hypothesis is supported: descriptor choice strongly affects nocturnal VPR performance.
2. GardensPoint remains the right primary benchmark because it directly matches the day-to-night problem.
3. The best descriptors we tested are now **SelaVPR++ and SALAD**, with CosPlace still a strong practical baseline.
4. The strict campus dataset is much harder than GardensPoint and exposes the remaining gap between benchmark success and deployment success.
5. VPRTempo did **not** transfer well here, mainly because the official checkpoint is a Nordland demo model rather than a universal descriptor.
6. The live system is already in place, so the next phase is to connect the strongest descriptors and lightweight temporal improvements to a robust demo.

What we would do next

- integrate **SALAD** and **SelaVPR++** into the live path as the strongest current descriptors
- test lightweight improvements such as **CLAHE** and **sequence matching**
- improve **no-match rejection** for campus deployment
- compare accuracy against runtime and deployment cost for a practical robot demo

Demo message

Our contribution is not only a benchmark comparison, but a practical descriptor study for night-time localization in a real campus environment.